Pixel-Level Assessment of Pavement Cracks With Light-Weight Convolution Network and Metric-Skeleton Measurement Using High-Resolution Images From Low-Cost Camera

Jin Wang, Tao Zhang, Hao Li, Niuqi Xu, and Yanyan Chen

***Abstract*— This study introduces a visual severity grading method for pavement cracks by using high-resolution images as inputs. The method incorporates a light-weight, fast crack segmentation network (CS-Net) to accurately extract cracks from images, and measure their lengths and widths with an image metric-skeleton model. Additionally, a five-level severity grading system is proposed to delineate the severity of crack lengths and widths. The CS-Net performs better than existing state-of-the-art networks on experiments, with $F_{1-score}$, intersection over union (IoU), frames per second (FPS), and model sizes of 78.66%, 65.25%, 55.00fps and 1.66MB, respectively. The metric-skeleton model precisely measures the crack with an overall mean error of approximately 0.30 pixels. According to maintenance guidelines, the distress grading level of pavement cracks is correspondingly assigned based on the length, width, and area values. The study findings can guide in allocating the maintenance budget for fixing highly defective road sections.**

***Index Terms*— Pavement crack, high-resolution image, road maintenance, machine learning, CS-Net, metric-skeleton measurement.**

I. Introduction

EXTRACTION and assessment of pavement cracks are crucial for road maintenance and ensuring driving safety. In practical road scenarios, crack widths and lengths are the most prevalent forms of pavement distress, serving as standard indicators to quantify the visual status of cracks. Figure 1 shows the example images of pavement distresses that suffer from crack issues. Conventional manual inspection methods of pavement cracks are typically time-consuming, subjective and labor-intensive [1]. Images and laser scanning are combined with a pavement management system, and help transport authorities prioritize rehabilitation work and optimize maintenance budgets by monitoring the condition of road networks over time [2], [3], [4]. However, it remains challenging to quickly process and detailed measure the pavement distress from high-resolution image data of road surfaces. Furthermore, although studies have focused on emerging machine learning methods to automatically detect and assess pavement distress [5], implementing pixel-level assessment remains challenging because of limited fast segmentation network and precise crack extent measurement from low-cost cameras. Therefore, the pixel-level assessment of pavement cracks by using low-cost images are significant research themes in intelligent road maintenance.

Recent studies have made considerable efforts to investigate deep learning-based models for extracting cracks using two-dimensional (2D) images [6], [7], [8], [9], [10], [11], However, high-cost computations and an increase in network parameters have limited the extension of deep learning networks in crack extraction. Meanwhile, most studies have focused on a single task (e.g. detection or segmentation), whereas limited studies have addressed pavement distress quantification and evaluation [5]. To address the high-cost computations associated with existing networks and achieve quantitative evaluation of individual cracks on asphalt pavements, in this study, a novel pixel-level crack status evaluation method is proposed by incorporating a light-weight convolution network and an image metric skeleton model with high-resolution images. This study makes three contributions:

(1) A light-weight crack segmentation network (CS-Net) is employed to successfully extract thin cracks from asphalt pavements using high-resolution images. The CS-Net achieves impressive fast performance while maintaining high accuracy in crack extraction. The effectiveness of the CS-Net is validated by a self-built dataset (asphalt pavement crack, APCrack600) and the publicly available EdmCrack600 dataset.

(2) A metric-skeleton model is proposed to delineate the lengths and widths of cracks in pixel-level from images. This module integrates Zhang's camera calibration algorithm and

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Fig. 1. Example images of asphalt pavement cracks.

a skeletonize operation to automatically measure the actual width and length of cracks from the image pixels. This experiment can be easily to perform on fields, and the mean deviations of crack widths is 2.81mm compared with manual measurements.

(3) The visual status of individual pavement cracks is assessed using low-cost camera images as inputs, and a five-level crack grading system from the pixel-level measurements of widths, lengths, and areas of cracks as outputs. The results adhere to road maintenance guidelines, mitigating the subjective evaluation of severity status and facilitating the allocation of the maintenance budget to the most highly defective road sections.

The remainder of this paper is organized as follows. Section II reviews the relevant literature, and Section III introduces the proposed method. Section IV details the experiments and reports the results performed using the proposed method. Finally, Section V summarizes the conclusions of the study and provides recommendations for future research.

II. Literature Review

A. Classification and Segmentation Networks in Image Processing

Deep learning has achieved notable breakthroughs in computer vision, and provides robust solutions to complex problems. Notably, it has demonstrated promising outcomes in the field of road crack detection. Early studies established the efficacy of a fully convolutional network (FCN) in effectively segmenting road cracks [7], [12]. Subsequently, a series of networks such as U-net [13], SegNet [14], DeepLab [15] and SegFormer [16] were proposed, and the structures of the networks have been optimized to improve networks crack segmentation results [9], [10], [17], [18], [19], [20], [21].

Despite the precision achieved by these networks, their efficiency and hardware demands are yet to be investigated comprehensively. For example, a MRA-Unet, which was developed from a U-net architecture [22], performs well in classification results but does not achieve the desired time consumption. Generally, these improvements of U-net, DeepLab, and SegNet provided promising results in crack segmentation; however, the problem of time consumption has not been concerned adequately. Furthermore, hardware consumption restricts quick response in decision-making, because road maintenance departments are typically allocated with resource-constrained hardware.

To overcome these challenges, studies have focused on the trade-off between time efficiency and network accuracy. Light-weight network architectures have been explored [23], [24], [25], [26], [27], [28], [29]. The light-weight and small model size of ENet is designed for semantic segmentation in autonomous driving scenes [30]. Meanwhile, depth-wise separable convolutions [31] and attention mechanisms [32] enhance inference speeds without losing accuracy. Recent models such as Segnext [33] introduce a multi-scale convolutional attention module, reducing parameter while maintaining high performance. Similarly, Unext [34], an MLP-based lightweight segmentation model, achieves computational efficiency with high segmentation accuracy in real-time applications with low computational complexity. However, the simplified architectures cannot satisfactorily perform tasks that require precise segmentation of fine details, which can result in a loss of granularity. Therefore, designing a novel crack segmentation network that can deliver excellent performance and ensure fast efficiency is desirable.

B. Crack Extent Measurement

Measuring the lengths and widths of cracks is a prerequisite to assess the severity of cracks. Tong et al. [35] applied deep convolutional neural networks (DCNN) to recognize the lengths of pavement cracks in batches. An image-based grid algorithm was developed to extract cracks [36]. A fast parallel thinning algorithm [37] was used to extract crack skeletons based on DeepLabV3+ [38], [39]. Meanwhile, a Laplace equation [40], an orthoBoundary algorithm [41], a binocular vision and SIFT feature point matching algorithms [42], and deep Q-learning network [43] have been developed to measure crack widths. However, these methods might be not precise, especially when dealing with images. Yang et al. [44] extracted crack skeletons and quantified the cracks in pixels. The measurement errors of crack lengths and widths are relatively large, ranging from −48.03% to 177.79%, and −13.27% to 24.01%, respectively. Analyses have revealed that the measurement errors are influenced by raw image resolutions or constrained by the metric measurement strategy.

The precision of crack measurements is contingent on the resolution of raw images; low-resolution images prove insufficient for detecting thin cracks, whereas high-resolution images obtained through professional vehicles are typically costly. In response, researchers are investigating the use of low-cost cameras for collecting road data. By using a low-cost camera mounted on a vehicle, Mei and Gül [45] proposed a EdmCrack600 dataset with high-resolution images. This study utilizes this dataset and conducts a new field experiment using a low-cost GoPro 11 camera to evaluate the performance of the network and investigate road application.

Regarding metric measurement methods, an ensemble of CNNs and a skeleton extraction algorithm were introduced to measure crack widths and lengths on images [46]. However, the measurement of cracks was based on an image-based coordinate system, the lengths and widths of cracks in real-world coordinate systems have not yet been determined. To establish the transformation between an image-based coordinate system and a real-world coordinate system, Zhang's camera calibration method [47] is widely used to accurately estimate camera parameters. While this method is simple and effective for image projection in computer vision applications, resolving

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the measurement of crack widths and lengths from the crack pixels to the actual cracks remains a challenge.

C. Assessment of Severity Level of Pavement Cracks

The evaluation of distress severity in pavement cracks is crucial for guiding road maintenance decision-making. A distress identification manual [48] categorizes cracking, patching and potholes, surface deformation, surface defects and miscellaneous distresses as primary factors influencing pavement distresses; longitudinal and transverse cracks are the most common among these factors. As per the "technical code of maintenance for urban road" (CJJ36-2016) [49], crack width measuring 10 mm or less should be repaired using specialized grouting (sealing) materials or hot asphalt; crack width exceeding 10 mm should be repaired following clause 5.3.7 of this code. In terms of evaluating crack widths, the long-term pavement performance (LTPP) distress identification manual [48] classifies crack width into three grading levels based on pavement width thresholds of 6 and 19 mm. Following these guidelines, precise assessments for guiding maintenance can be obtained by measuring the widths and lengths of cracks. However, to the best of the authors' knowledge, no study has established a measurable mathematical relationships between raw images and the distress status of the widths and lengths of individual pavement cracks based on maintenance guidelines.

III. Methodology

The proposed methodology consists of three components: a CS-Net for extracting pixel-level pavement cracks from high-resolution images, measuring crack widths and lengths with a metric-skeleton model, and grading the evaluation of asphalt pavements with respect to crack widths, lengths and areas.

A. CS-Net for Extracting Pavement Cracks From High-Resolution Images

The light-weight encoder-decoder based CS-Net (Figure 2), inspired by U-net [13] and ENet [30], comprises of three main blocks: FeatureFirst block, FeatureDown block, and FeatureUp block [30].

When an input image with green-red-blue (RGB) information is put into the network, it undergoes feature initialization through the FeatureFirst module. Following this, encoding and decoding are predominantly accomplished using three pairs of FeatureDown and FeatureUp modules. The indices preserved by the FeatureDown module are specifically designed to minimize hardware demands for upsampling. In addition to the three main blocks, the CS-net incorporates a conv2d block, a dilated block, and a separable block are used in the CS-Net [30] (Figure 3). The FeatureFirst, FeatureDown and FeatureUp modules are explained in the following paragraphs.

To capture the morphological characteristics of cracks, the FeatureFirst block (Figure 4) is devised to augment the number of feature learning channels using separable convolutions in the initial stage of CS-Net. This operation effectively captures the geometric features of cracks and considerably reduces the number of convolution parameters. It integrates standard and separable convolutions in RGB images. Multi-scale feature information with four branches is connected through concatenation. The weights of these four branches are automatically adjusted during the network training stage. The Conv2d operation in the diagram takes as input the number of input channels, number of output channels, convolutional kernel size, and stride. As illustrated in Figure 4, the input feature map size is set to height$\times$ width $\times$ 3 ($H \times W \times 3$), and the four dimensions of the output are designed as 3, 5, 3, 5. Consequently, a feature map with a size of ($H \times W \times 16$) is produced.

The FeatureDown block (Figure 5) is designed to reduce the spatial resolution of the feature maps and preserve important spatial information. This block is divided into two main branches. The left branch executes max-pooling with a 2×2 convolutional kernel and a stride of 2, concurrently outputting the corresponding index information (Indices). The right branch establishes an attention mechanism by applying average pooling and max pooling along the channel dimension, along with a 7×7 convolutional kernel. An inverted residual module that combines standard and separable convolutions is then employed to capture both local and global contextual information through three additional convolutional operations. The final step involves feature fusion derived from the outputs of the two branches.

The FeatureUp block (Figure 6) adapted from the Enet structure [30] is used in the upsampling stage. The low-resolution feature maps are transformed into high-resolution features while integrating features from the downsampling path. This block also incorporates information from the downsampling path. Subsequently, the indices preserved by the FeatureDown block generate a new feature map that is equivalent to the resolution of the input image.

A loss function that measures the overlap between the predicted results and true labels allows the model to predict crack boundaries more accurately. The dice loss function [50] helps the network focusing on the foreground area. However, because of the imbalance in the number of samples between non-crack pavement surfaces and pavement cracks, it is often difficult to extract thin cracks. Meanwhile, the focal loss function [51] is introduced to combine the boundary characteristics of the cracks. Therefore, the final loss function is a weighted combination of the dice and focal loss functions.

B. Pixel-Level Measurement of Crack Widths and Lengths Using a Metric-Skeleton Model

The metric-skeleton model (Figure 7) including Zhang's camera calibration algorithm [47] and a skeletonize operation is developed to accurately measure the cracks from image pixels to millimeter measurements in real-world road scenarios.

By inputting the pavement cracks from the CS-Net, the flexible Zhang's camera calibration method estimates the camera intrinsic matrix ($\mathbf{M} = \begin{bmatrix} f_x & 0 & u_0 & 0 \\ 0 & f_y & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$) and the extrinsic parameters (rotation matrix $\mathbf{R}$ and translation vectors $\vec{t}$), where (f_X, f_y) is the focal length in the camera coordinate system $(\mathbf{U}, \mathbf{V})$, and (u_0, v_0) is the geometric center of the image in

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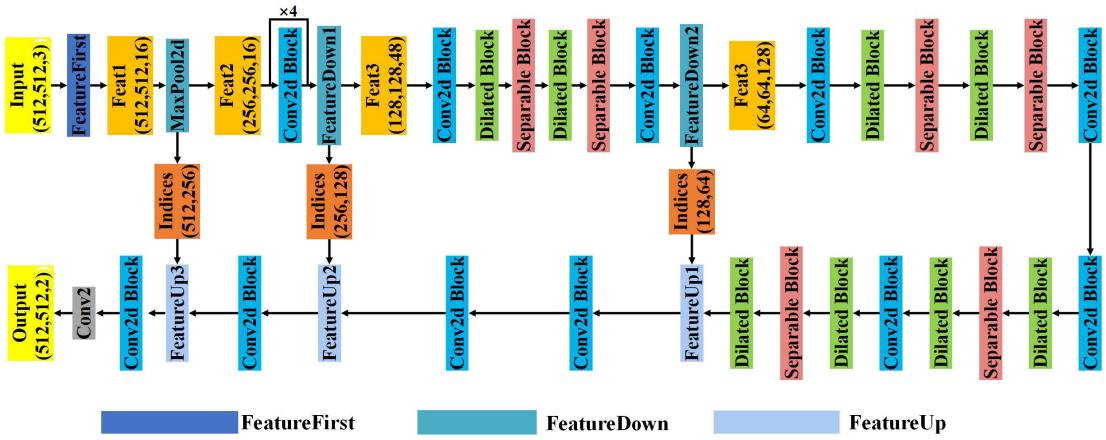

Fig. 2. The overview architecture of the CS-Net.

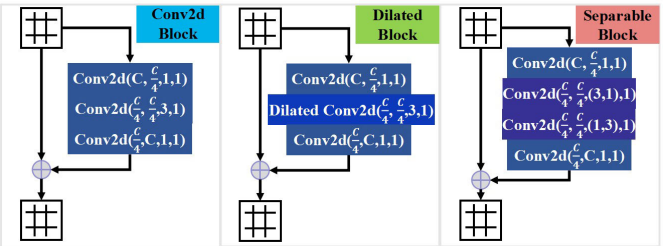

Fig. 3. Structures of a conv2d block (left), a dilated block (middle), and a separable block (right).

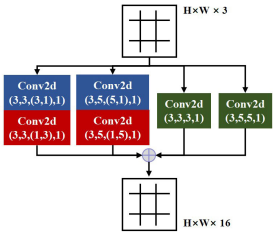

Fig. 4. Workflow of the FeatureFirst block.

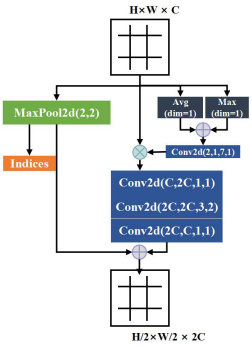

Fig. 5. Workflow of the FeatureDown block.

the camera coordinate system ($\mathbf{X_c}, \mathbf{Y_c}, \mathbf{Z_c}$). The transformation function from the real-world coordinate system ($\mathbf{X_w}, \mathbf{Y_w}, \mathbf{Z_w}$) to the camera coordinate system is expressed as follows [47]:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{M} \begin{bmatrix} \mathbf{R} & \vec{t} \\ \vec{0} & 1 \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix} \tag{1}$$

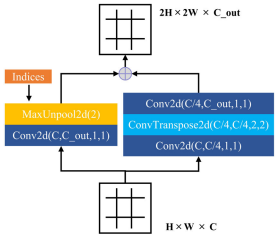

Fig. 6. Workflow of the FeatureUp block.

where s is the arbitrary scale factor, (u, v) are the pixel coordinates of the images, and (x_w, y_w, z_w) are the pixel coordinates in the real-world coordinate system. Because the distances between the camera mounted on the vehicle and the pavement surface were stable in the experiment, in this study, $z_w = 0$. Therefore, the scale factor (s) is expressed as follows:

$$s = (\mathbf{R}^{-1}\vec{t})_{[2,0]} / (\mathbf{R}^{-1}\mathbf{M}^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix})_{[2,0]} \tag{2}$$

The intrinsic and extrinsic parameters of the camera are determined using a closed-form solution, followed by non-linear refinement based on the maximum likelihood criterion [47]. Consequently, each pixel in an image can be measured in metric units.

To evaluate the width of an individual crack (right one on Figure 7), a skeletonization operation is performed. This operation involves segment-based normal vector estimation [37] and a predefined proportion of crack widths. Assuming n segments exist in an individual crack, the distance between each pair of skeleton points ($(x_i^l, y_i^l, 0)$ and $(x_i^r, y_i^r, 0)$) in the segment is computed to represent the width of the corresponding crack segment. For individual cracks, all segment widths are re-aligned in descending order, with larger widths of road segments with a predefined proportion being used to represent the widths of individual crack. The width of an individual crack (L_w) is expressed using the following Equation:

$$L_w = \frac{1}{|\mathbb{C}|} \sum_{i \in \mathbb{C}} \sqrt{(x_i^l - x_i^r)^2 + (y_i^l - y_i^r)^2},$$
$$\mathbb{C} = \left\{ c_j \mid c_j \in \mathbb{A}, min(\mathbb{C}) \geq max(\mathbb{A} \setminus \mathbb{C}), |\mathbb{C}| = \lceil 0.3n \rceil \right\} \tag{3}$$

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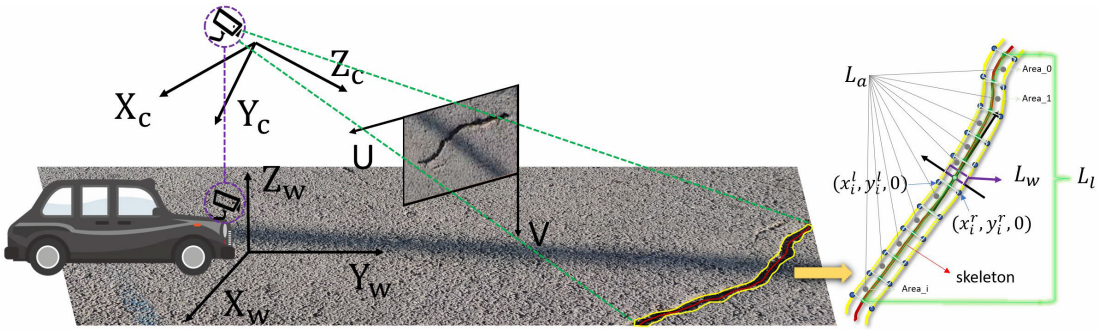

Fig. 7. Estimating crack lengths and widths based on Zhang's camera calibration (left) and the skeletonize operation (right).

where n represents the total number of the crack-skeleton pairs. Here, $\mathbb{C}$ is the set of all skeleton points corresponding to the first 30% of the width in the set $\mathbb{A}$; and c_j is the element at index j in the set $\mathbb{A}$. The length of the individual crack (L_l) is represented by the sum of the lengths of all segments, where the length of each segment is computed using the start and end pixels of the skeleton points on the segment. Subsequently, the crack area (L_a) is computed by multiplying the width and the length of the crack.

C. Width-Length-Based Severity Assessment of Asphalt Pavement Cracks

Using the measured pavement cracks, the severity levels of the cracks are assessed based on the estimated lengths (L_l), widths (L_w), and areas (L_a) of cracks. When the crack width is less than 10 mm, CJJ36-2016 recommends the use of specialized sealant material or hot asphalt sealant. The LTPP distress identification manual classifies cracks into three levels based on the crack width: $\leq$ 6 mm as low; $>$ 6 mm and $\leq$ 19 mm as moderate; and $>$ 19 mm as high. Although the severity level is specified mostly based on the crack width, in this study, more quantifying factors (length, width, and area) are integrated into the severity grading system to finely describe crack extents. Therefore, a novel five-level severity grading system is established to precisely evaluate the distress status of individual cracks (Table I). The lowest threshold of 0.006 m^2 corresponds to a crack width of 6 mm and a length of 1 m; the threshold of 0.010 m^2 corresponds to a crack width of 10 mm and a length 1 m; the threshold of 0.019 m^2 corresponds to a crack width of 19 mm and a length 1 m; and the upper threshold of 0.038 m^2 corresponds to a crack width of 19 mm and a length of 2 m. When the area $\leq$ 0.006 m^2, severity level I is assigned; when 0.006 $<$ area $\leq$ 0.010 m^2, severity level II is assigned; when 0.010 $<$ area $\leq$ 0.019 m^2, severity level III is assigned; when 0.019 $<$ area $\leq$ 0.038 m^2, severity level IV is assigned; and when the area $>$ 0.038 m^2, severity level V is assigned.

IV. Experiments and Analysis

A. Datasets and Experiment

1) EdmCrack600: The publicly available EdmCrack600 [45] is used to test the performance of CS-Net. The EdmCrack600 consists of 600 images with dimensions of 1920 $\times$ 1080 pixels. It includes pavement distresses such as transverse cracks and longitudinal cracks. For more details, please refer to [45].

TABLE I

Five-Level Distress Grading System of Pavement Cracks From Image Data

Severity Level of Pavement Cracks	Crack Width (L_w) $/mm$	Crack Length (L_l) $/m$	Crack Area (L_a) $/m^2$
Level I	≤ 6	≤ 1	≤ 0.006
Level II	≤ 6	> 1	$0.006 < \& \leq 0.010$
	$6 < \& \leq 10$	≤ 1	
Level III	$6 < \& \leq 10$	> 1	$0.010 < \& \leq 0.019$
	$10 < \& \leq 19$	≤ 1	
Level IV	$10 < \& \leq 19$	> 1	$0.019 < \& \leq 0.038$
	>19	≤ 2	
Level V	>19	> 2	> 0.038

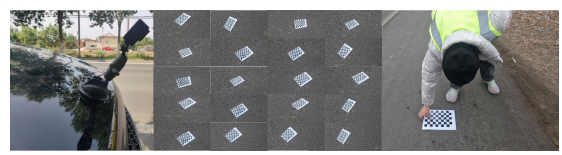

Fig. 8. APCrack600 dataset collected by a GoPro 11 camera mounted on the vehicle and the chess-board field experiment.

2) APCrack600: A field experiment (Figure 8) was performed in Pinggu District, Beijing, China. A GoPro 11 camera was mounted on the vehicle hood, and one planar chess-board pattern of A4 size was placed on the pavement surface in front of the vehicle. The position of the pattern was manually adjusted during the process to obtain relative positions between the camera and the pattern. The vehicle speed was approximately 20 km/h. The video's resolution was 3840 $\times$ 2160 pixels at 30 fps. The camera calibration includes performing camera calibration on a flat surface, fixing the camera to the vehicle hood, and adjusting the camera's viewing angle for external parameter calibration. The tested asphalt road with a length of approximately 20 km mainly consisted of thin transverse and longitudinal cracks. Figure 9 demonstrates the pavement images and corresponding annotated images, where the red color denotes the annotated cracks. The thin cracks ranged within 3-60 mm. This dataset, named APCrack600, consisted of 600 images with a size of 3840 $\times$ 2160 pixels. Pixel-wise annotations were performed on each crack image using LabelMe software [52]. The image samples were grouped into two categories: pavement cracks and others.

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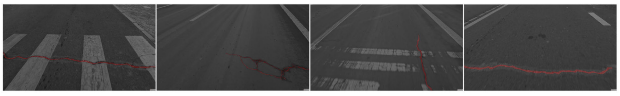

Fig. 9. Sample annotations of cracks in the APCrack600 dataset.

B. Implementation Details and Evaluation Metrics of the CS-Net

The segmentation networks were implemented using Windows 10 on a single graphics processing unit with an NVIDIA GeForce RTX 3070 (16GB RAM). Our training central processing unit was an 11th Gen Intel Core i7-11800H @ 2.30GHz. The code was tested using PyTorch 1.13.1, CUDA 11.6, and Python 3.7.16. The learning rate lr_{max} and lr_{min} were set to 1e-2 and 1e-4, respectively. A total of 300 epochs and eight batch sizes were used during the training stage to converge the model. To reproduce the experimental results, the global random seed was set to 20. Data augmentation (random resizing and distortion, horizontal flip, and color space transformation) was performed to prevent overfitting and improve network accuracy. The training, validation, and testing samples were divided in a ratio of 7:1:2 to ensure the robustness of the model.

The mean intersection over union (MIoU) is used as a process metric to monitor the overall progress of networks. Crack intersection over union (IoU), mean pixel accuracy (MPA), and $F_{1-score}$ were applied to evaluate the quality of crack segmentation.

C. Network Comparisons in Processing Efficiency

To compare the time efficiency and hardware requirements, state-of-the-art networks (FCN [53], U-net [13], PSPNet [54], DeepLabV3 [55], SegNet [14], ENet [30], BiSeNet [56], ICNet [57]), SegFormer [16], Unext [34] and Segnext [33] were compared in Table II on the EdmCrack600 dataset. FLOPs was estimated from an input of $3 \times 512 \times 512$. The parameters were calculated by the thop.profile tool. The model size originates directly from the actual size of the trained weight files in our experiments, which reflects the true size of saved weights after training. The proposed CS-Net achieved a better overall performance than the state-of-the-arts networks, with FLOPS, parameters, model size, inference time, and FPS values of 3.02, 0.35, 1.66, 18.18, and 55.00, respectively. In our experiment, ResNet 50 was applied as the basic encoder module of U-net, DeepLabV3, PSPNet, and ICNet. The ENet was stored in fp16 format. Meanwhile, the standard ResNet18 was applied as the based model of BiSeNet, and the real-time MiT-B0 encoder variant was used in SegFormer. They cause the parameters and model sizes larger than that of the results given in the literature. The CS-Net achieved moderate results between computational complexity and model size with FLOPs and model parameters of 3.02 G, and 0.35 M, respectively. The FPS score for CS-Net was 55.00 on the NVIDIA GeForce RTX 3070 with an input image size of 1920×1080. This illustrates that CS-Net provides a promising solution for fast crack segmentation applications.

D. Ablation Experiment

An ablation test of loss function was conducted on the EdmCrack600 dataset (Table III). By combining the dice loss and focal loss function, the IoU and $F_{1-score}$ achieved better results than that of the dice loss alone.

To assess the effectiveness of the FeatureFirst and FeatureDown modules in CS-Net, ablation experiments were conducted on the public dataset EdmCrack600. The experimental results are presented in Table IV. Because the proposed CS-Net is partly inspired by ENet, the corresponding modules from ENet were used to replace the proposed FeatureFirst and FeatureDown modules in the ablation experiments. In Table IV, "-FF" represents the use of the initialization module from ENet instead of the FeatureFirst module, while "-FD" represents the use of the downsampling module from ENet instead of the FeatureDown module. The CS-Net-FF, at the initial input stage of the network, leads to unnecessary emphasis on less meaningful features from the local self-attention in the FeatureDown module. The proposed FeatureDown module effectively enhanced the network's learning ability.

E. Extraction Results of Pavement Cracks on EdmCrack600 and APCrack600 Datasets

For the EdmCrack600 dataset, BiSeNet, DeepLabV3, ENet, FCN, ICNet, PSPNet, SegNet, U-net, SegFormer, Unext, Segnext and CS-Net were compared, and the validation process on the training stage are presented in Figure 10. The convergence of the loss function on Figure 11 illustrates that the CS-Net learns the crack features and adapts to the crack segmentation task effectively. Table V and Figure 13 detail the evaluation results of crack extraction among FCN, U-net, PSPNet, DeepLabV3, SegNet, ENet, BiSeNet, ICNet, SegFormer and the proposed CS-Net. The CS-Net achieved superior overall performance, which shows IoU, Recall, Precision, MPA, and $F_{1-score}$ of 46.82%, 64.72%, 62.87%, 82.23%, and 63.78%, respectively. Enet is designed for real-time segmentation on automatic driving scenes, and is suitable for multi-class segmentation. On this dataset, the imbalance samples between the cracks and the background cause the low performance of Enet. Meanwhile, the structure of SegNet attracts more detailed features in its lower layers, which can reduce false positives. It may cause the precision of CS-Net to be lower than that of SegNet. As shown on Figure 13, the CS-Net captures more details of thin cracks and learns more neighboring features of individual cracks. In this dataset, fewer longitudinal crack samples existed compared with transverse samples. This imbalance influenced the ability of the model to generalize and segment longitudinal cracks effectively, which can result in poorer results, as depicted on Figure 12.

For the APCrack600 dataset, a comparison was conducted between BiSeNet, DeepLabV3, ENet, FCN, ICNet, PSPNet, SegNet, U-net, SegFormer, Unext, Segnext and CS-Net; the validation results are presented in Figure 14. The results indicate that CS-Net achieved high-quality segmentation results during the validation stage. The convergence of the loss function illustrates that CS-Net effectively addressed the crack segmentation task on APCrack 600 (Figure 11). Table VI and

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TABLE II

PERFORMANCE COMPARISONS ON THE EDMCRACK600 DATASET AMONG THE CS-NET AND OTHER NETWORKS (FLOPS ESTIMATED FROM AN INPUT OF $3 \times 512 \times 512$)

Model	FCN	U-net	PSPNet	DeepLabV3	SegNet	ENet	BiSeNet	ICNet	SegFormer	Unext	Segnext	CS-Net
FLOPs/G	155.66	276.64	184.73	171.25	160.68	**2.18**	13.05	36.97	6.76	2.30	28.78	**3.02**
Parameters/M	35.43	28.85	46.71	39.76	29.44	**0.35**	12.80	26.24	3.71	1.47	18.50	**0.35**
Model Size/MB	143.31	110.12	186.34	168.84	112.44	**1.58**	51.21	108.26	14.24	5.65	70.9	**1.66**
Inference Time/ms	32.51	77.97	39.52	35.36	36.97	25.69	**10.47**	20.07	35.28	21.76	37.94	**18.18**
FPS	30.76	12.83	25.30	28.28	27.05	38.92	**95.53**	49.82	28.35	45.96	26.36	**55.00**

TABLE III

COMPARISON AMONG THE LOSS FUNCTIONS FOR CS-NET ON EDMCRACK600 DATASET

Loss Function	IoU/%	$F_{1-score}$/%
Dice	46.19	63.19
Dice + Focal	46.82	63.78

TABLE IV

ABLATION ANALYSIS OF CS-NET ON EDMCRAC600 DATASET

Model	IoU/%	Recall/%	Precision/%	MPA/%	$F_{1-score}$/%
CS-Net - FF	0	0	0	0	0
CS-Net - FD	38.85	51.33	61.51	75.53	55.96
CS-Net	**46.82**	**64.72**	**62.87**	**82.23**	**63.78**

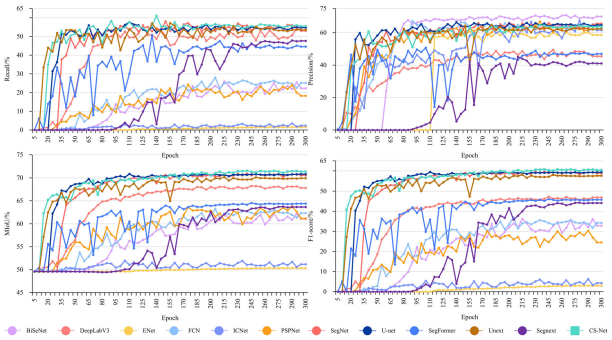

Fig. 10. Comparisons of the evaluation results of EdmCrack600 validation dataset.

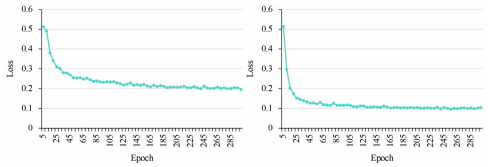

Fig. 11. Loss function curves of EdmCrack600 (left) and APCrack600 (right).

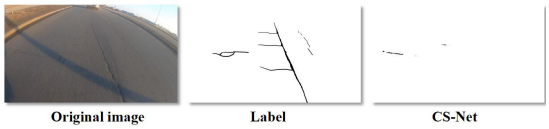

Fig. 12. The failed prediction case of CS-Net on EdmCrack600.

TABLE V

COMPARISONS OF EVALUATION RESULTS OF EDMCRACK600 DATASET

Model	IoU/%	Recall/%	Precision/%	MPA/%	$F_{1-score}$/%
FCN	24.99	23.68	54.88	51.12	30.01
U-net	44.47	60.36	62.83	80.04	61.57
PSPNet	21.23	19.12	62.57	50.77	26.50
DeepLabV3	36.82	47.62	44.32	52.28	42.78
SegNet	44.55	57.97	**65.81**	78.87	61.64
ENet	1.36	1.38	60.58	50.69	2.70
BiSeNet	23.10	21.69	49.24	50.01	27.52
ICNet	5.40	3.23	39.75	50.02	5.25
SegFormer	29.80	46.76	45.11	73.15	45.92
Unext	43.23	60.40	60.34	80.05	60.37
Segnext	27.18	45.66	40.17	72.57	42.74
CS-Net	**46.82**	**64.72**	62.87	**82.23**	**63.78**

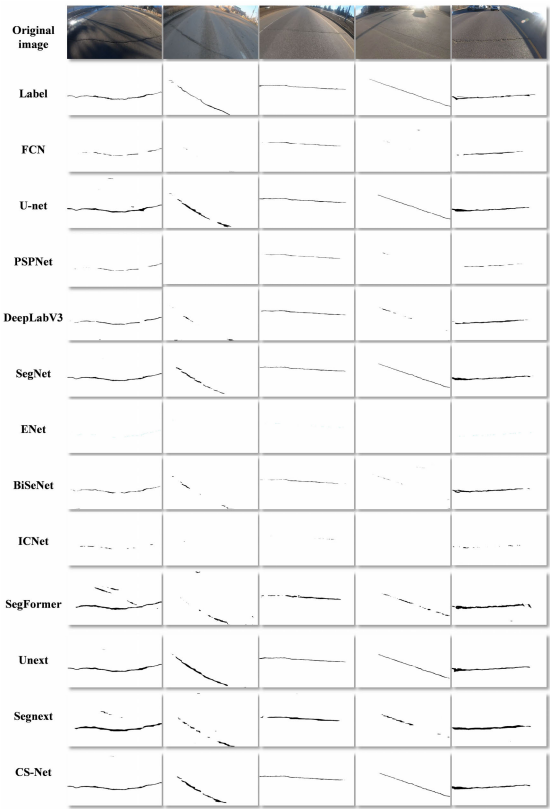

Fig. 13. Examples of the evaluation results of EdmCrack600 dataset.

Figure 15 showcase the evaluation results of crack extraction using FCN, U-net, PSPNet, DeepLabV3, SegNet, ENet, BiSeNet, ICNet, SegFormer, Unext, Segnext and CS-Net. The proposed CS-Net exhibited superior overall performance with MIoU, Recall, Precision, MPA, and $F_{1-score}$ scores of 65.25%, 82.75%, 75.52%, 91.42%, and 78.66%, respectively. Compared with the EdmCrack600 dataset, the Enet exhibits effectiveness on the APcrack600 dataset because of the various setting of data collection (camera angle and lens settings). In our APCrack600 dataset, the camera was tilted downward and the images consists of pavement surfaces mostly. Meanwhile, a linear field-of-view mode was used to reduce the phenomenon of fisheye distortion. CS-Net demonstrated a high level of consistency in identifying the continuity of pavement cracks and accurately captured the fine-grained details of cracks. Compared with other networks, the widths of

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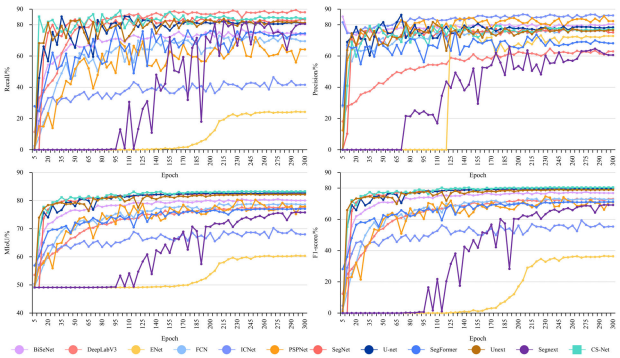

Fig. 14. Comparisons of the evaluation results of APCrack600 validation dataset.

TABLE VI

COMPARISONS OF EVALUATION RESULTS OF APCRACK600 DATASET

Model	IoU/%	Recall/%	Precision/%	MPA/%	$F_{1-score}$/%
FCN	31.55	34.25	**81.00**	56.23	43.84
U-net	63.54	78.65	76.78	89.12	77.70
PSPNet	6.62	7.61	79.64	50.40	12.44
DeepLabV3	41.91	54.85	69.03	61.63	55.10
SegNet	63.19	80.64	74.49	90.08	77.44
ENet	20.18	22.12	69.69	60.98	33.58
BiSeNet	31.01	34.43	80.10	54.40	43.98
ICNet	19.37	21.42	79.90	54.09	30.50
SegFormer	54.85	73.62	68.27	86.50	70.85
Unext	64.18	82.08	74.64	90.80	78.18
Segnext	52.45	78.30	61.38	88.72	68.81
CS-Net	**65.25**	**82.75**	75.52	**91.42**	**78.66**

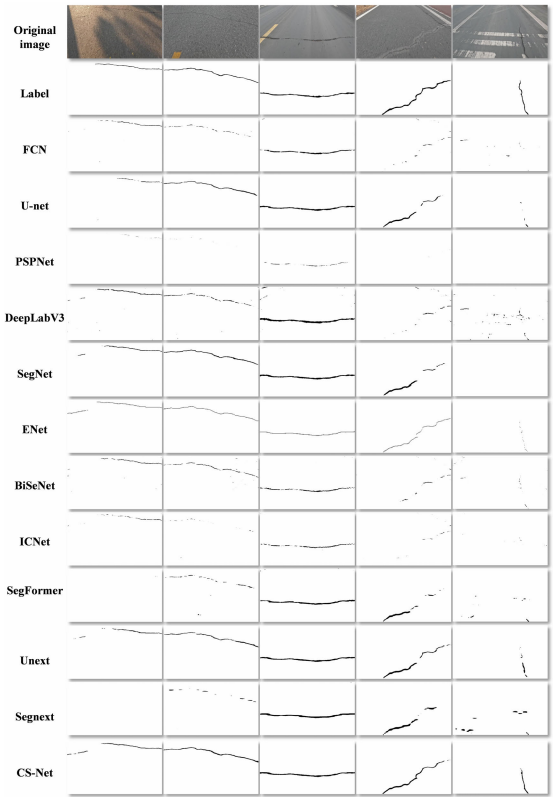

Fig. 15. Examples of the evaluation results of APCrack600 dataset.

the extracted cracks revealed closer results with ground truth labels.

The proposed CS-Net provides a promising solution for fast performance and good-quality model outputs on both EdmCrack600 and APCrack600 datasets. Its competitive performance in terms of quality and efficiency indicates that the CS-Net is a reasonable choice for measuring road cracks.

TABLE VII

ESTIMATED PARAMETERS OF ZHANG'S CAMERA CALIBRATION

Camera Parameters	Estimated Results
(f_x, f_y)	(2001.2902,2001.3826)
(u_0, v_0)	(1904.8560,1043.7058)
R	$\begin{bmatrix} -0.9978 & 0.0576 & 0.0328 \\ -0.0101 & -0.6218 & 0.7831 \\ 0.0656 & 0.7810 & 0.6210 \end{bmatrix}$
t	[178.3729 405.6344 1174.4105]

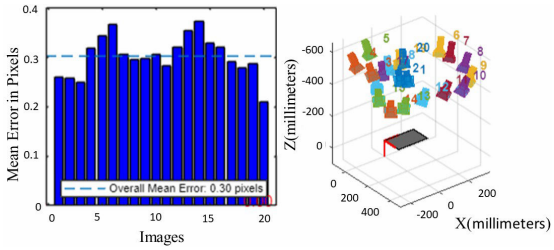

Fig. 16. Reprojection errors and visualization of camera pose.

F. Measurement Results of Crack Lengths and Widths

On the self-built APCrack600 dataset, 20 chess-boards were used to determine the internal and external parameters of the camera in the field experiment. Table VII lists the internal and external parameters of camera calibration, and Figure 16 shows the projection errors and visualization of the camera poses in the experiment. The overall mean error and the maximum error were 0.30 pixels and 0.37 pixels, respectively. According to [58], when the overall mean error was less than 0.50 pixels, the precision satisfied the quality of the measured crack widths and lengths.

Table VIII presents the sample comparisons of the crack lengths and widths between the proposed method and manually measured results. Manual observations were conducted using a steel tape in the experiment. The deviations in crack lengths fluctuated within −7.8 to 296.7 mm, with estimated crack lengths of 488.9 and 2203.3 mm, respectively. The deviations in crack widths ranged from −17.9 to 1.1 mm, with estimated crack widths of 13.1 and 62.9 mm, respectively. The average relative deviations in the crack lengths and widths from C1 to C9 were 3.8% and 15.1%, respectively. The largest deviations appeared on cracks C8 and C9 primarily because of the image-blurring phenomenon caused by the high driving speeds of the vehicle. Although errors were observed in the CS-Net and metric-skeleton measurements, these errors had limited influence on measuring the widths and lengths of pavement cracks.

G. Severity Grading-Level Results of Pavement Cracks

As listed in Table I, the severity grading-level results of pavement cracks on the APCrack600 dataset are summarized and illustrated in Figure 17. Among 117 images exhibiting crack issues, 30 individual cracks were classified as Level I,

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TABLE VIII
CRACK LENGTH AND WIDTH COMPARISONS BETWEEN PREDICTED AND MANUALLY MEASURED RESULTS

Crack Number	Crack Lengths (mm)				Crack Widths (mm)			
	Manual	Prediction	Deviation	Relative Deviation (%)	Manual	Prediction	Deviation	Relative Deviation (%)
C1	530.0	537.8	**-7.8**	1.5	5.5	5.8	-0.3	5.5
C2	578.0	567.0	11.0	1.9	3.0	2.0	1.0	33.3
C3	495.0	488.9	6.1	1.2	13.0	13.1	-0.1	0.8
C4	487.0	473.6	13.4	2.8	10.0	9.9	0.1	1.0
C5	495.0	479.1	15.9	3.2	5.5	5.1	0.4	7.3
C6	429.0	427.1	1.9	0.4	3.5	3.3	0.2	5.7
C7	2500.0	2203.3	**296.7**	11.9	25.0	23.9	**1.1**	4.4
C8	2500.0	2289.4	210.6	8.4	11.0	15.2	-4.2	38.2
C9	2700.0	2622.1	77.9	2.9	45.0	62.9	**-17.9**	39.8
Average				3.8	Absolute Deviation		2.81	15.1

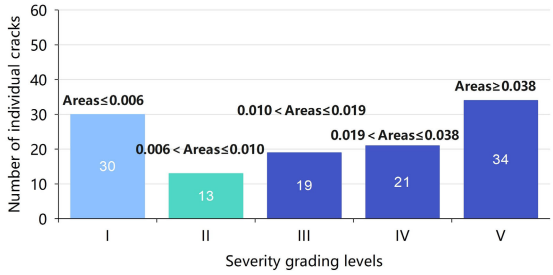

Fig. 17. Severity grading levels of pavement cracks on the testing dataset of APCrack600.

13 as Level II, 19 as Level III, 21 as Level IV, and 34 as Level V. These severity grading results provide quantitative measurements for assessing the visual status of pavement cracks, facilitating precise decision-making of road maintenance applications.

V. CONCLUSION

A pixel-level severity grading method, comprising CS-Net and metric-skeleton measurement, was proposed to quantitatively evaluate the pavement crack lengths and widths using high-resolution images from a low-cost camera. Because of classical image-based deep learning methods have struggled to balance between efficiency and output quality, the light-weight CS-Net designed with the new FeatureFirst and FeatureDown blocks and fewer model parameters. It achieved fast efficiency and high performance quality compared with state-of-the-art networks (BiSeNet, DeepLabV3, ENet, FCN, ICNet, PSPNet, SegNet, U-net and SegFormer) on two datasets, the publicity available EdmCrack600 and the self-built APCrack600. The proposed metric-skeleton measurement model transforms image pixels into a real-world coordinate system and measured crack widths and lengths using pairs of segmented skeleton points in individual cracks. The overall mean error of the model was 0.30 pixels, satisfying the precision requirements for crack measurement. The severity grading results supported the quantitative assessment of pavement cracks consistent with road maintenance manuals. This study provides a precise guide for agencies in allocating maintenance budgets independently of labor-intensive field measurements. The method is well-suited for the fast visual inspection of pavement cracks by using a low-cost camera.

Future studies should focus on assessing a more comprehensive assessment of pavement distress, such as potholes and rutting. A quantification analysis on more image frames from the camera videos, as well as additional testing on various road types (e.g., expressway and rural roads) will be conducted in future. The CS-Net can be improved, especially in enhancing feature extraction for detecting smaller cracks in complex backgrounds. Metric-skeleton crack measurement should be further investigated for sub-pixel estimation of crack widths and lengths.

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